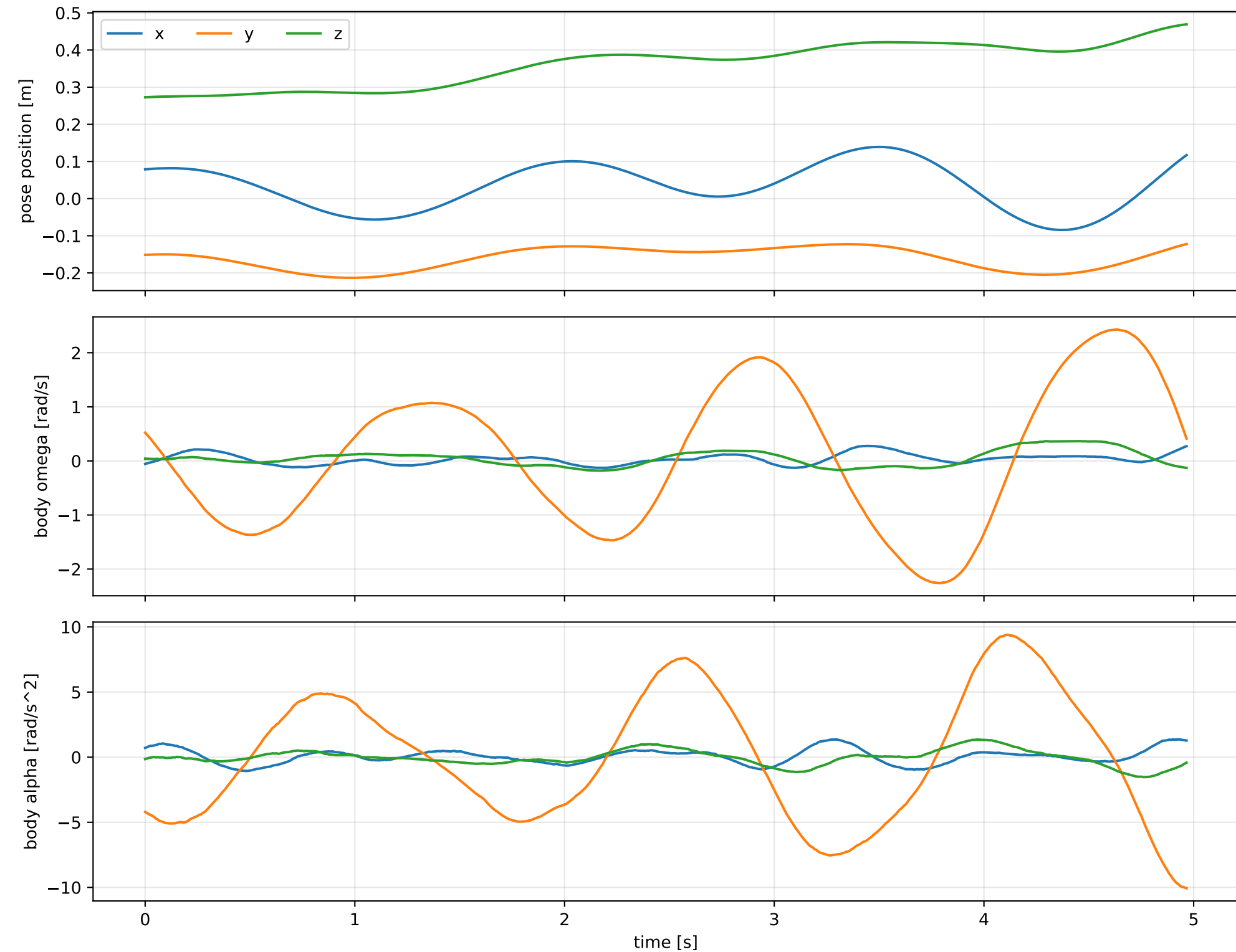
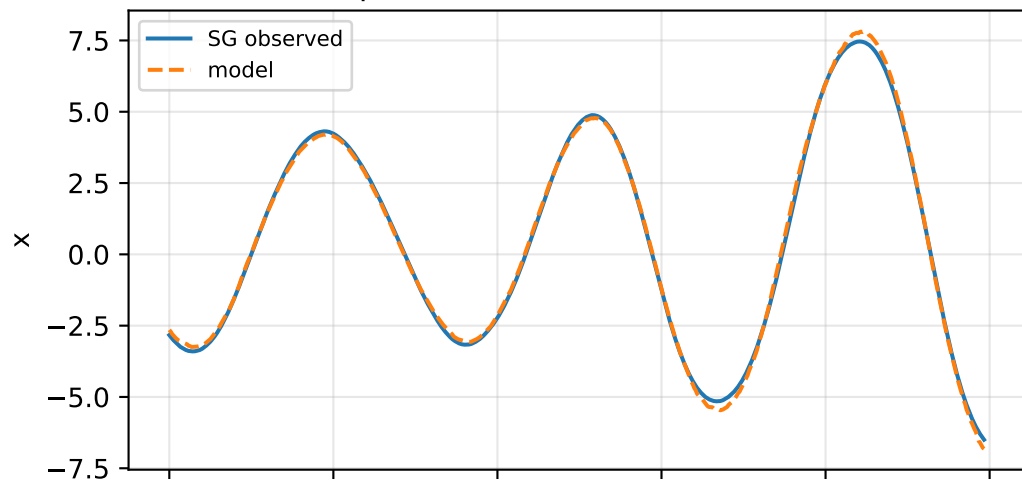
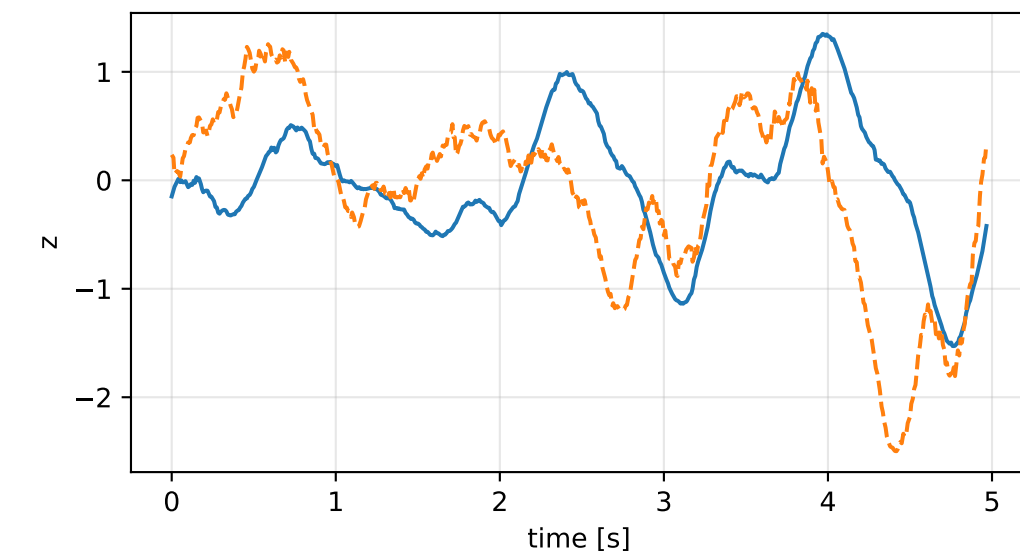
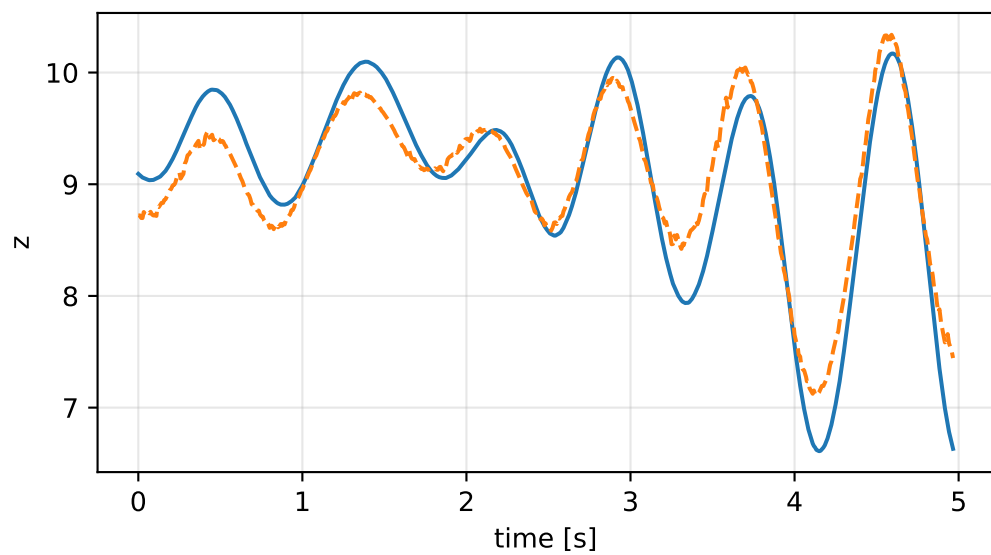
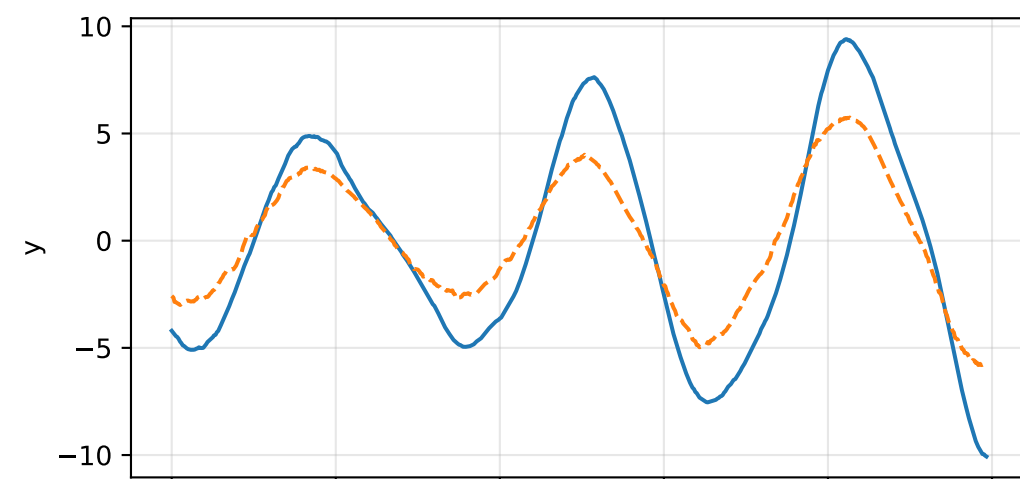
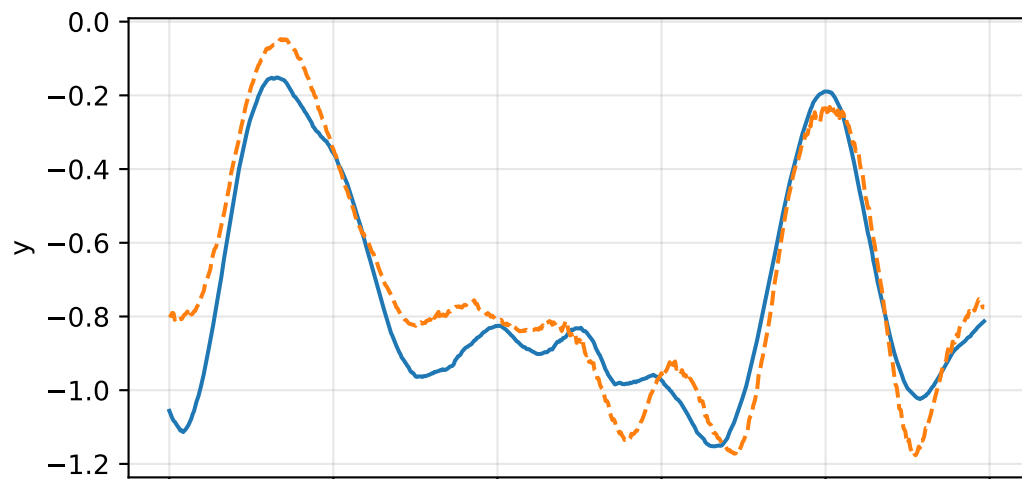
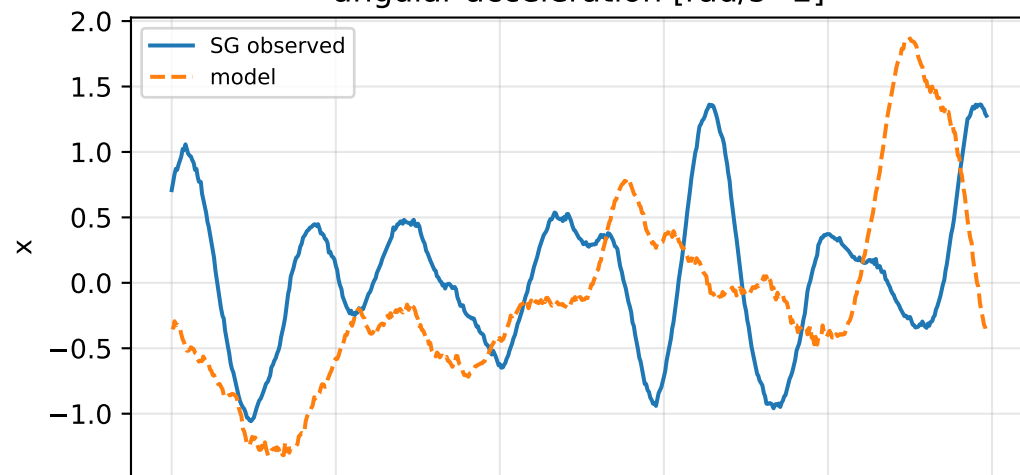


cov_full_no_R_uncertainty_in_s: SG trajectory and derived motion

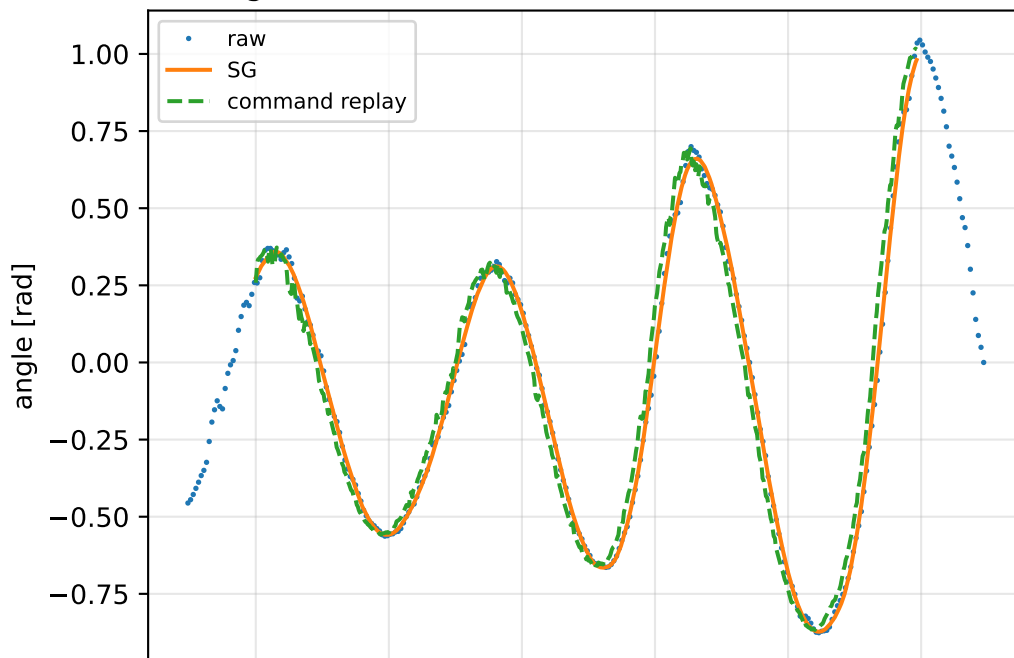


cov_full_no_R_uncertainty_in_s: acceleration residual objective

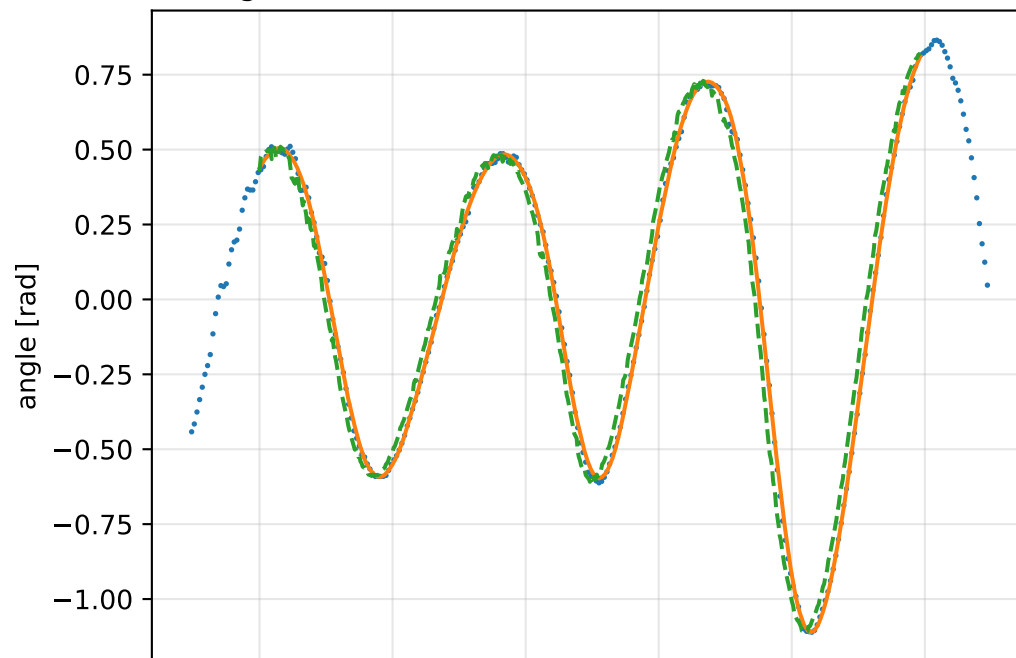
specific acceleration [m/s²]angular acceleration [rad/s²]

cov_full_no_R_uncertainty_in_s: gimbal measurement audit (objective=measured_sg)

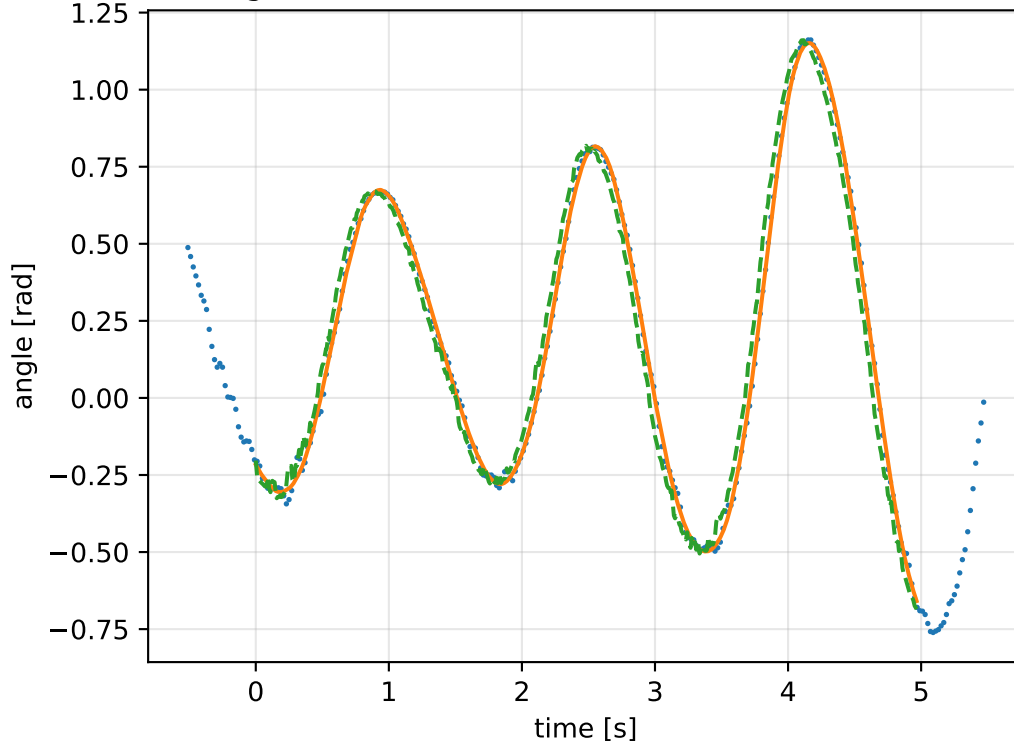
gimbal 1: RMSE=0.0759, max=0.2095 rad



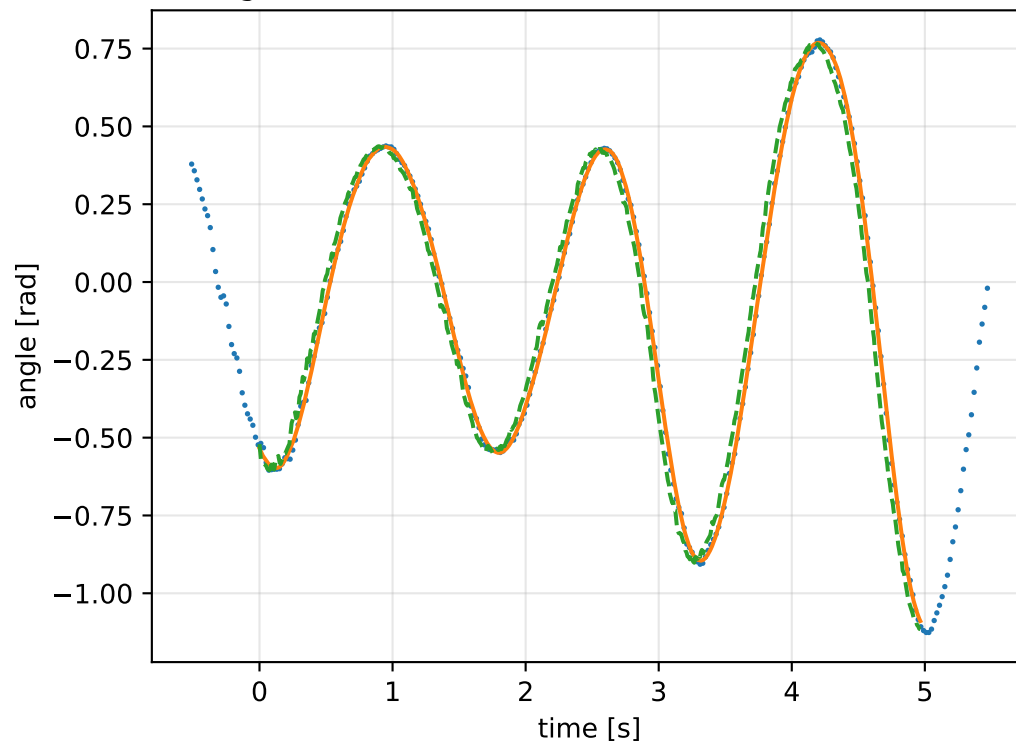
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

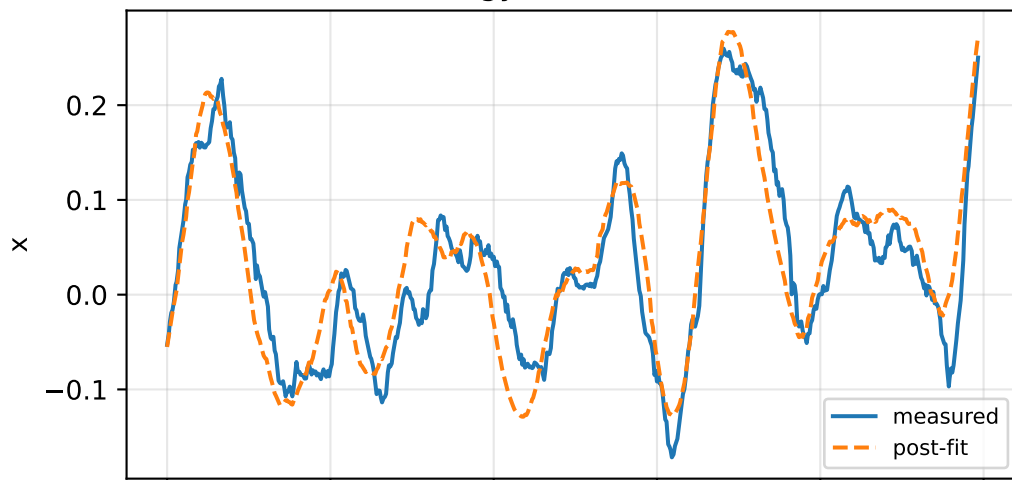


gimbal 4: RMSE=0.07131, max=0.1737 rad

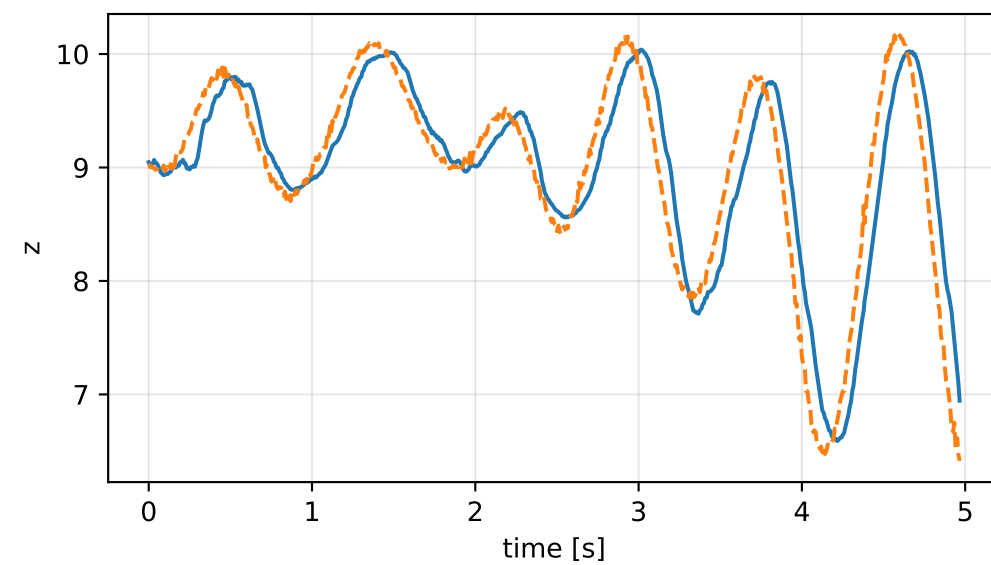
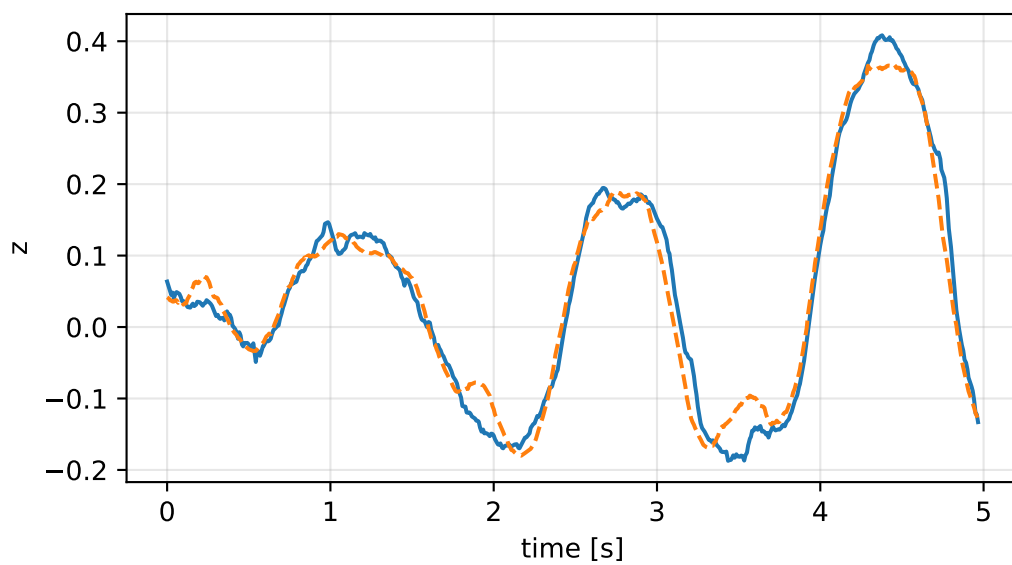
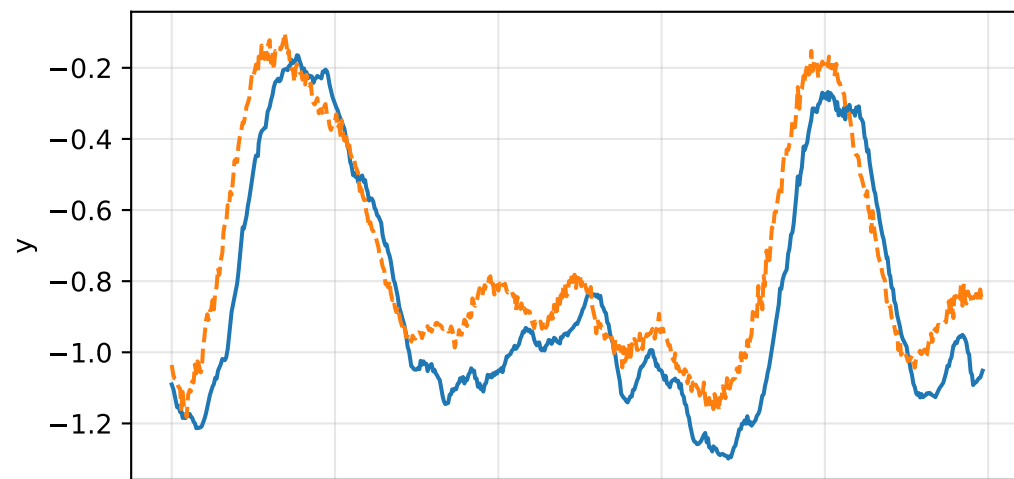
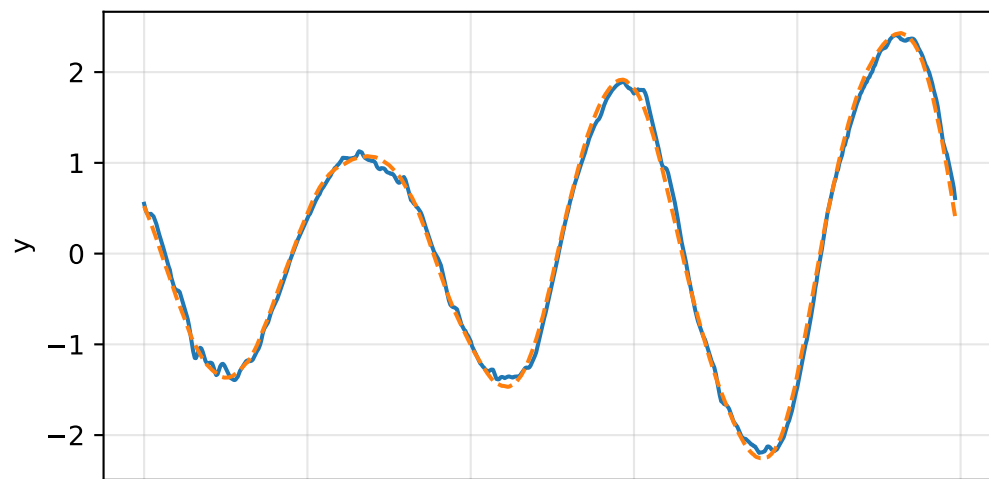
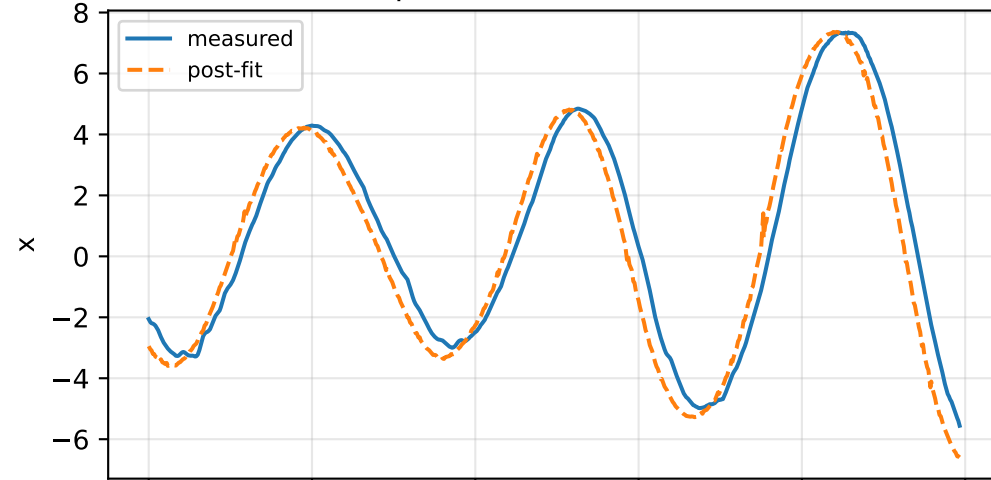


cov_full_no_R_uncertainty_in_s: IMU comparison (diagnostic only)

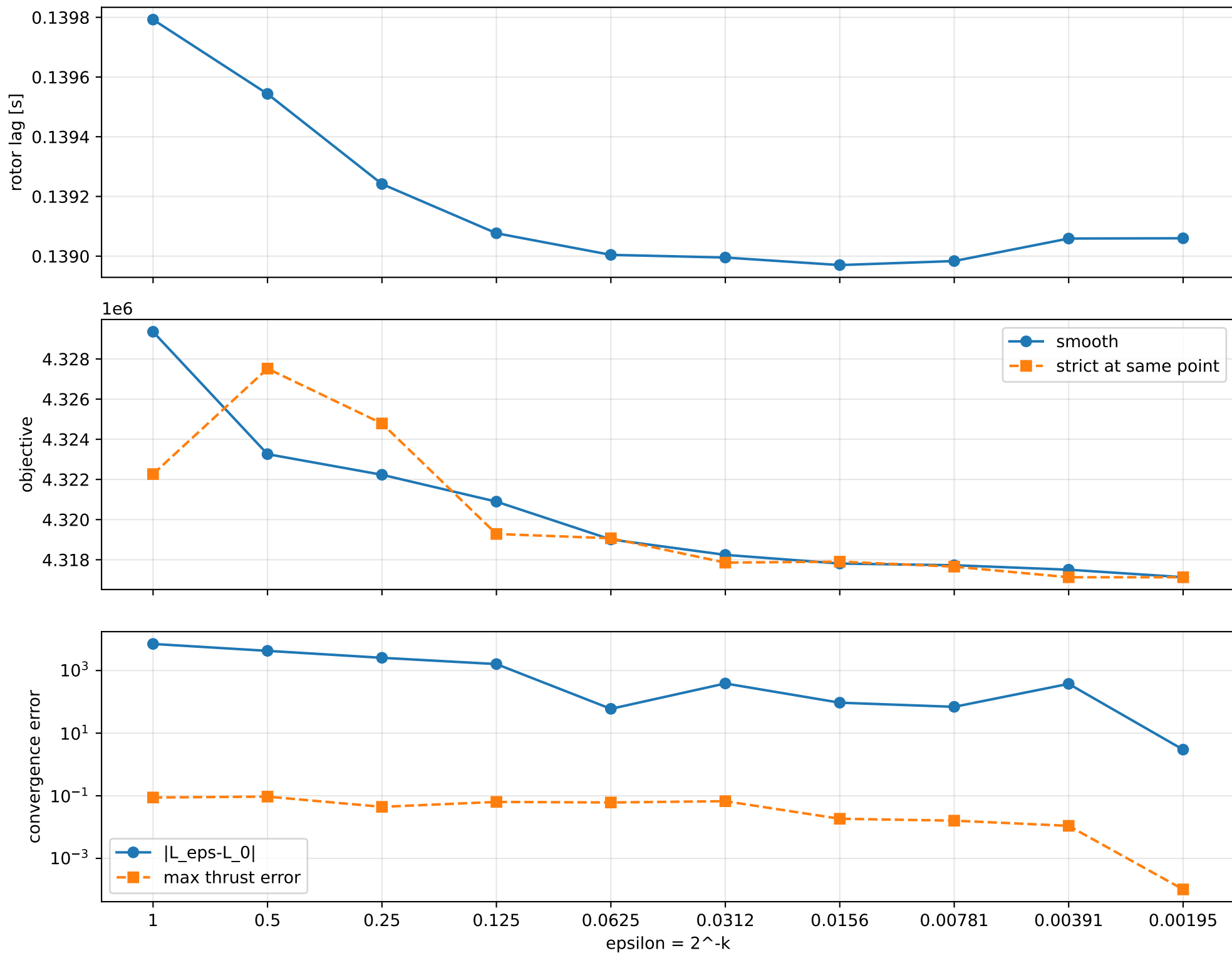
gyro [rad/s]



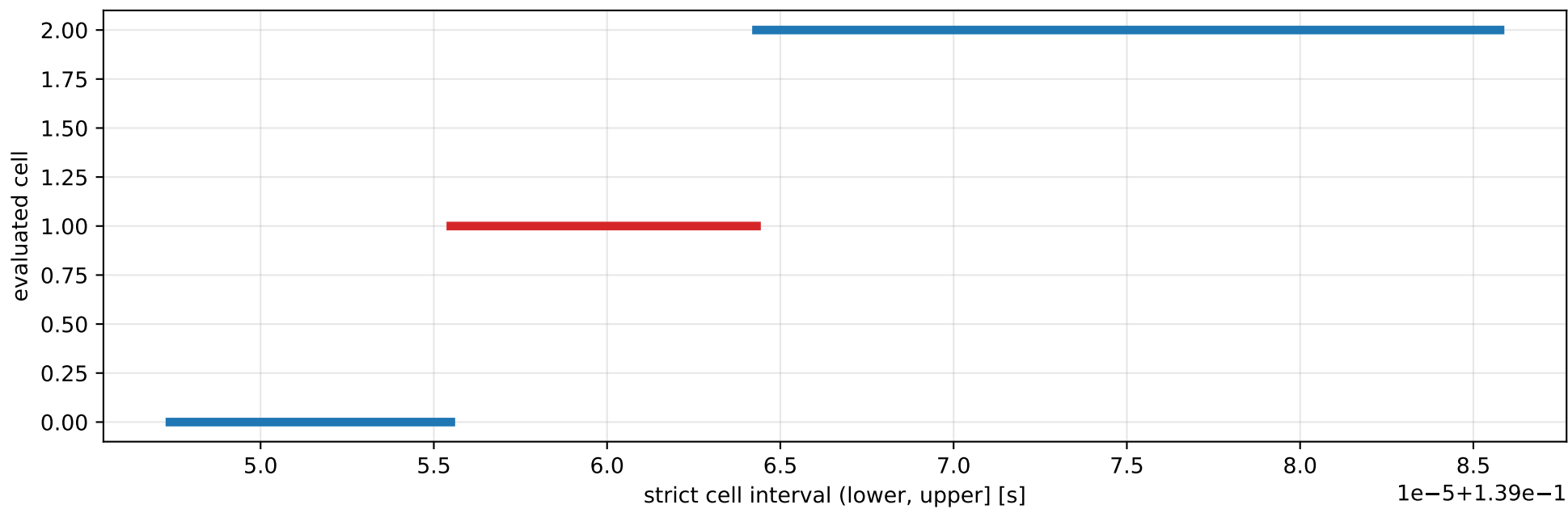
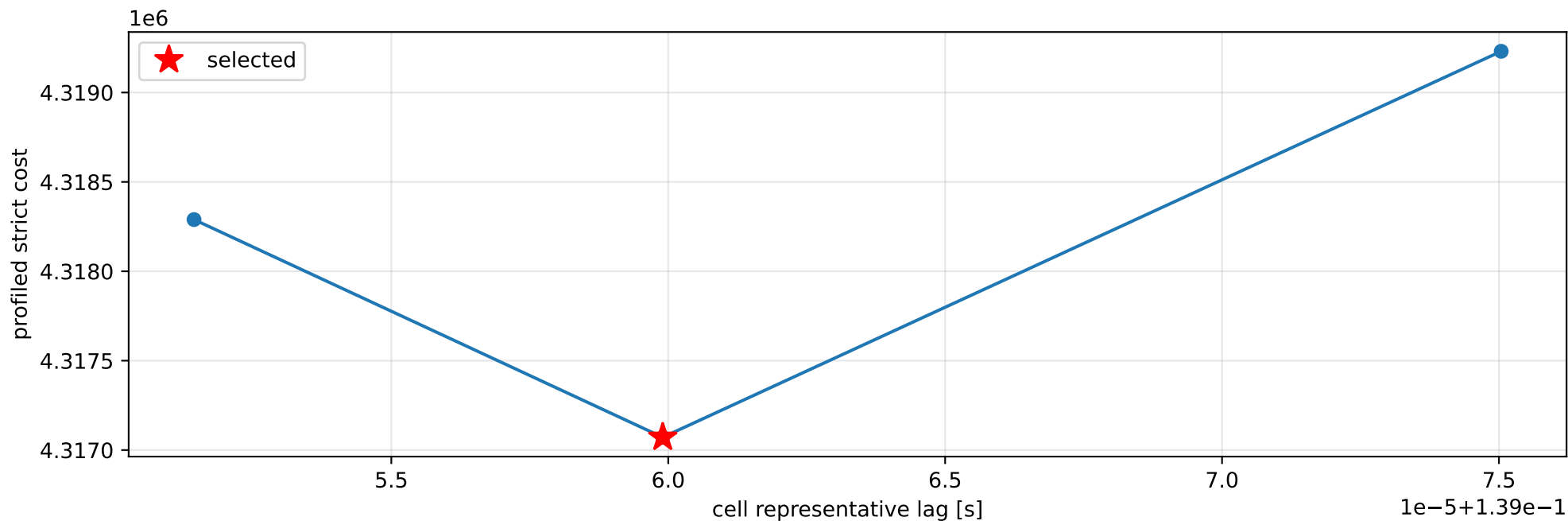
specific force [m/s²]



cov_full_no_R_uncertainty_in_s: smooth-to-strict continuation



cov_full_no_R_uncertainty_in_s: exact strict-ZOH lag cells



selected (0.13905549, 0.139064312] s; final smooth 0.139059856 s; data boundary=False

cov_full_no_R_uncertainty_in_s: parameter estimate

Case: cov_full_no_R_uncertainty_in_s
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.30409094

inertia [kg m²]:

```
[[ 0.1675474  0.00184456  0.05979077]
 [ 0.00184456  0.19452525 -0.03134259]
 [ 0.05979077 -0.03134259  0.08606209]]
```

CoG body [m]: [0.01334383 0.05909932 -0.06223408]

force effectiveness: [0.34364714 0.43601895 0.5997701 0.52930259]

scale-free J/m [m²]:

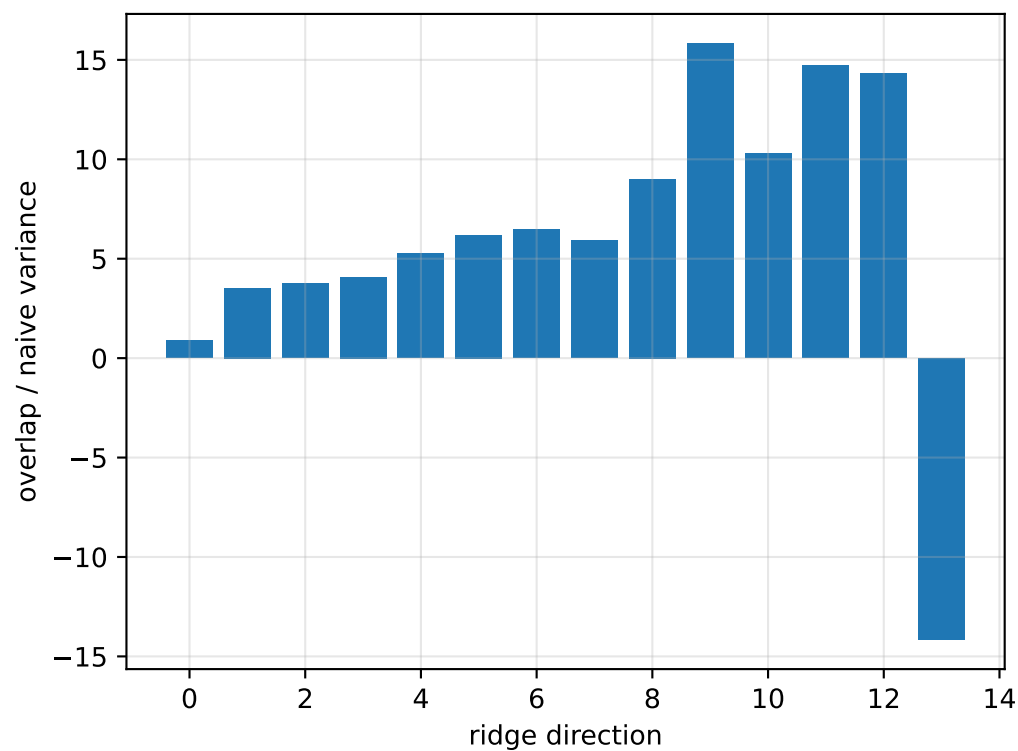
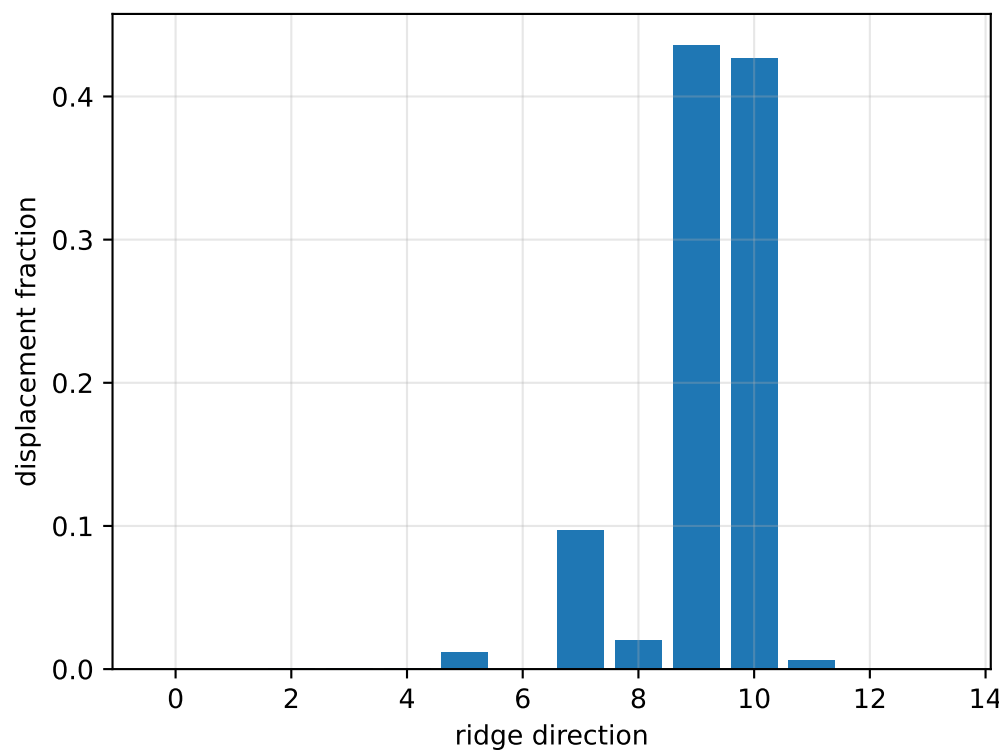
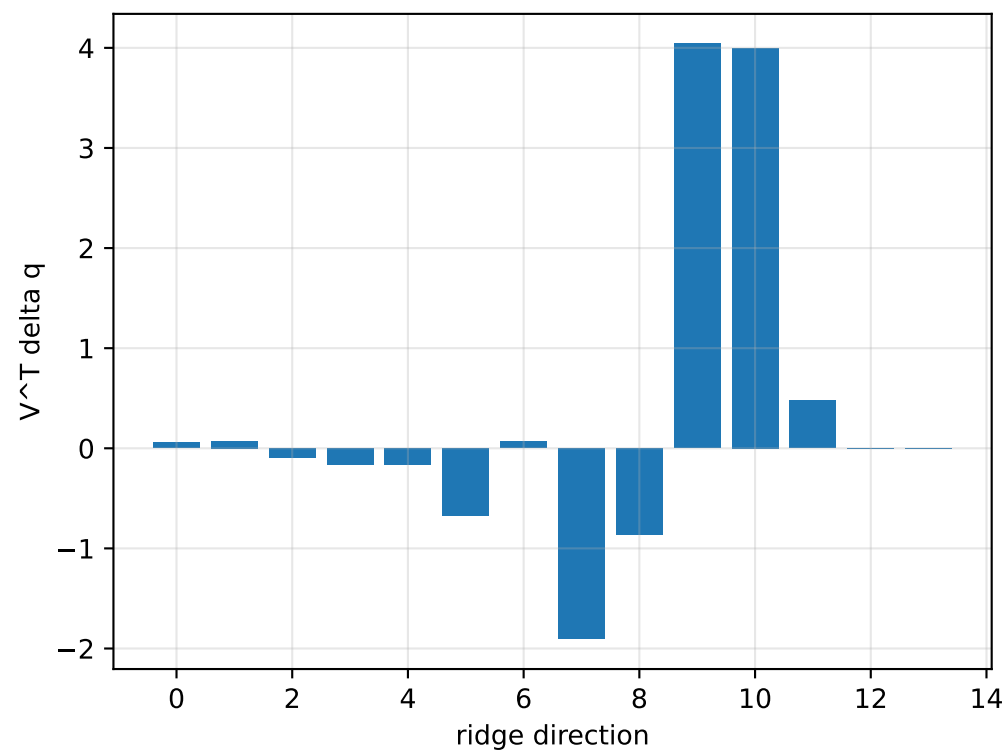
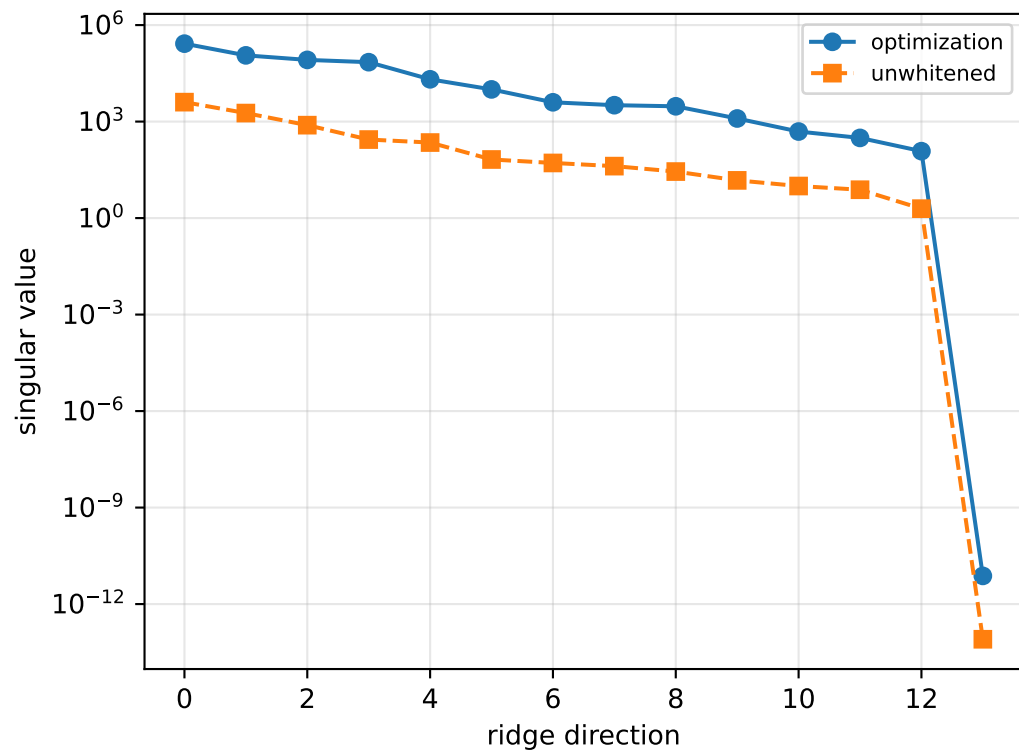
```
[[ 0.12847831  0.00141444  0.04584862]
 [ 0.00141444  0.1491654  -0.02403405]
 [ 0.04584862 -0.02403405  0.06599393]]
```

scale-free f/m: [0.26351471 0.33434705 0.45991432 0.40587859]

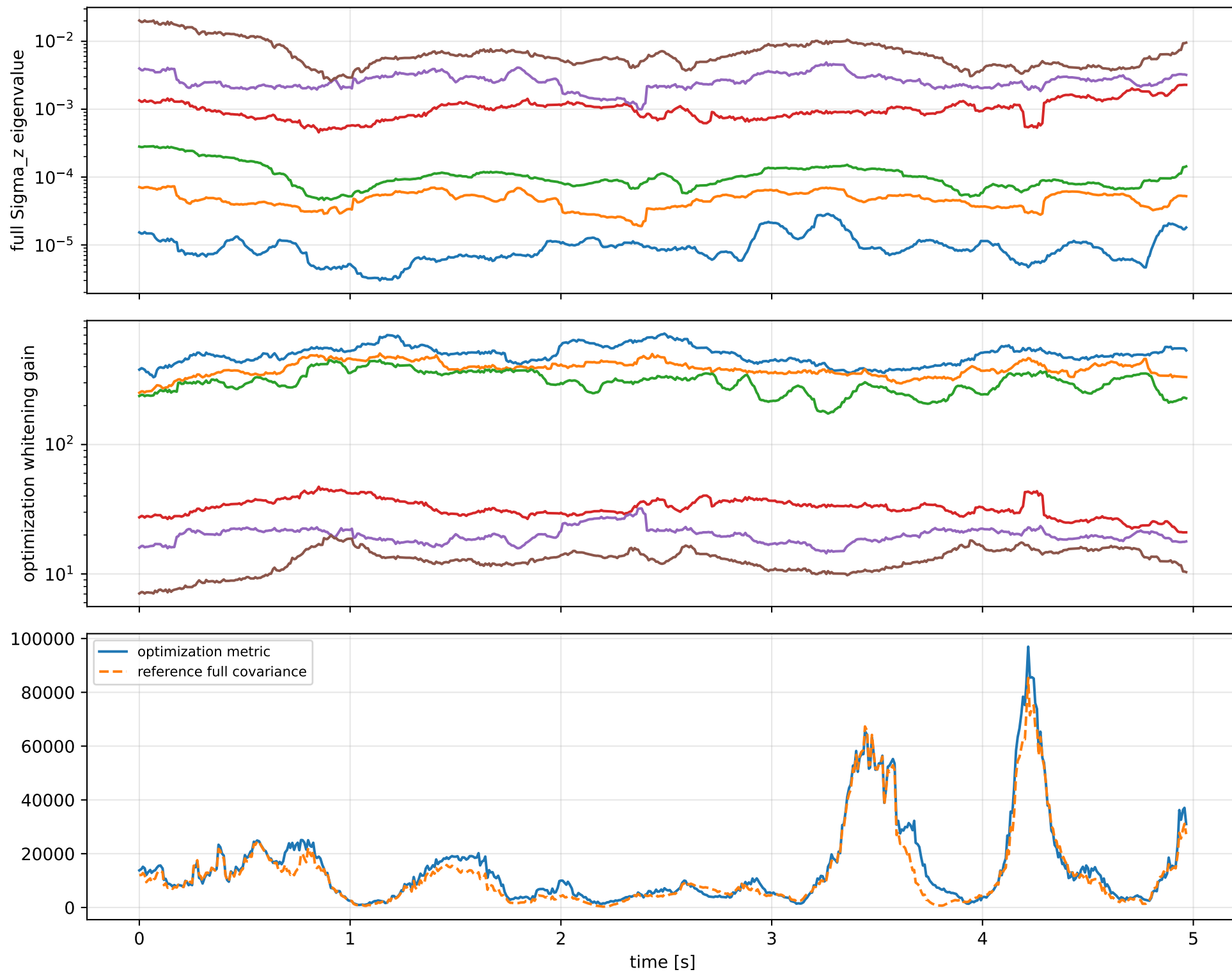
rotor lag cell: (0.13905549, 0.139064312] s

representative: 0.139059901 s

cov_full_no_R_uncertainty_in_s: ridge spectrum (rank 13, nullity 1, ||jv||=1.75e-11)



cov_full_no_R_uncertainty_in_s: optimization vs reference covariance



cov_full_no_R_uncertainty_in_s: wrench / closure (force max $4.83\text{e-}15$, torque max $3.89\text{e-}16$)

